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System reliability of a manufacturing network with reworking action and different failure rates

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From the perspective of network analysis, the manufacturing system can be constructed as a stochastic-flow network, since the capacity of each machine is stochastic (i.e. multistate) owing to the failure, partial failure, and maintenance. Considering reworking action and different failure rates of machines, the input flow (raw materials/work in process) processed by each machine might be defective, and therefore the output flow (work in process/products) would be less than the input amount. To evaluate the capability of the manufacturing system, we measure the probability that the manufacturing network can satisfy demand. Such a probability is defined as the system reliability. A decomposition method is first proposed to divide the manufacturing network into one general processing path and one reworking path. Subsequently, two algorithms are utilised for different network models to generate the lower boundary vector of machine capacity to guarantee that the manufacturing network is able to produce sufficient products fulfilling the demand. The system reliability of the manufacturing network can be derived in terms of such a capacity vector afterwards.

Keywords: manufacturing network; system reliability; reworking; multistate; different failure rates; decomposition method

1. Introduction

In the industry, it is a crucial task to evaluate the capacity of a manufacturing system for determining whether it can satisfy customers' orders/demand or not. From the decision-making and production management perspectives, the system reliability is one of the essential performance indicators to assess the capability of the manufacturing system (Black and Mejabi 1995, Savsar 2000, Adamyan and He 2002, Lin 2007a, Liu and Li 2009). The system reliability is the ability of a system to perform its intended function (Stevenson 2007). For a manufacturing system, such a performance indicator provides information of overall capability that a system could be.

1.1 *Network analysis for manufacturing systems*

To evaluate the performance of a manufacturing system, network analysis is an achievable tool to be applied. That is, the manufacturing system can be modelled as a manufacturing network (Lee and Garcia-Diaz 1993, 1996, Lin 1998, Lin 2007a, 2009, Listeş 2007, Francas and Minner 2009) and thus further analysis can be implemented based on such a network. Several studies (Lee and Garcia-Diaz 1993, 1996, Listeş 2007, Paquet *et al*. 2008, Francas and Minner 2009) have been devoted to performance measurement for a manufacturing system and supply chain by adopting network analysis. Lee and Garcia-Diaz (1993, 1996) applied a flow-network approach to solve the grouping problem in a manufacturing system by measuring the functional similarity between machines as a performance indicator. Listeş (2007) designed a generic stochastic model for a supply-and-return network in a closed loop system to locate the general manufacturing path and reworking path. Paquet *et al*. (2008) proposed an optimisation methodology to design the manufacturing network producing several products, in which the capacities of processors and workers are considered. Francas and Minner (2009) utilised the manufacturing network configuration in a supply chain with product rework to investigate the cost and profit under different network structures.

However, the above studies did not emphasise the capability for a manufacturing network with considering stochastic (multistate) capacities of each machine. Those works mainly focused on costs, profits, and sales for evaluating the performance of the manufacturing network. The capability of a manufacturing system reflects the

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order satisfaction ability directly. Nevertheless, evaluations of costs, profits, and sales mainly provide the financial information rather than the production performance. From the production-management perspective, this paper works on measuring the capacity of machines to determine if the manufacturing system satisfies the customers' orders or not.

1.2 *Failure and rework in manufacturing systems*

The failure rate of each machine influences the capability of a manufacturing system and would lead to defective products, in which defective products would be reworked or scrapped. Thus, an important issue to be considered is how the reworking action affects the amount of output products in a manufacturing system. In many cases, defective products still have substantial values, e.g. caused by expensive input materials, and thus it is an incentive to rework those products to an 'as new' condition (Buscher and Lindner 2007, Li *et al*. 2009). In several applications, the reworking action is implemented on the same machines for producing products. This implies that a manufacturing network would have two sources from general processing path and reworking path for satisfying demand (Buscher and Lindner 2007, Teunter *et al*. 2008).

1.3 *Stochastic-flow network*

In a manufacturing network, each arc can be regarded as a machine, and each node is an inspection station after the machine. Owing to the failure, partial failure, and maintenance, each machine in the manufacturing network has stochastic capacity levels (i.e. multistate) with performance ranging from a perfectly working level to complete malfunction. Therefore, the manufacturing network is also a multistate, and we can treat it as the so-called stochastic-flow network (Xue 1985, Lin *et al*. 1995, Lin 1998, 2003, 2004, 2007a, b, 2009, 2010, Zuo *et al*. 2007, Yeh 2008).

To measure the capability of a manufacturing network satisfying the customers' requirement, Lin (2007a) focused on the two-commodity reliability evaluation of a manufacturing network in terms of a minimal path (MP), in which an MP is a path the proper subsets of which are no longer paths. In Lin's work, the system reliability is defined as the probability that the manufacturing network satisfies two-commodity demand. A great deal of research (Xue 1985, Lin *et al*. 1995, Lin 1998, 2004, 2007a, 2009, 2010, Zuo *et al*. 2007, Yeh 2008) has also been devoted to studying the system reliability of a stochastic-flow network (such as manufacturing network, computer network, communication network, etc.) in terms of MP. In these studies, the demand transmitted through a network must obey the flow conservation (Ford and Fulkerson 1962), which implies that no flow will be increased or decreased during transmission. More importantly, some properties related to the manufacturing network, such as rework and scraps, were not considered in these studies.

For a practical manufacturing network, the input flow processed by each machine would not be the same as output flow, since the failure rate of each machine is considered. That is, the output products of the manufacturing network might be less than the input raw materials. It implies that the traditional methodology for the stochastic-flow network problem could not be applied in such a case owing to the disobedience of the flow conservation. Moreover, based on the MP concept, an arc (machine) would not appear on the same path more than once; otherwise, it would not be an MP. However, defective work in process (WIP) from a machine would be reworked, starting from previous machine(s) or the same machine(s) (Buscher and Lindner 2007, Teunter *et al*. 2008), which is breaking the basic concept of MP. To conquer these limitations, we propose a graphical method to revise the manufacturing processes for constructing the manufacturing network first. Based on the graph, the manufacturing network can be decomposed into several paths, an individual general processing path and a reworking path.

1.4 *Focusing problem*

Considering rework case, this paper utilises an activity-on-arrow network diagram approach to transform a manufacturing system into a manufacturing network. Based on the transformed manufacturing network, this paper evaluates the probability that a manufacturing network could produce d units of products with reworking action by applying a stochastic-flow network model. Such a probability is named system reliability herein. We consider each machine in the manufacturing network has stochastic capacity (i.e. multistate) and a failure rate. The flow in a

manufacturing network is defined as the input amount that each machine processes per unit time. Thus, the general processing path and reworking path are utilised to analyse the different input flows processed by each machine.

Two algorithms are proposed to generate the lower boundary vector, composed of the capacity for each machine, which ensures enough products satisfying demand d are produced. In terms of such a vector, the system reliability of the manufacturing network is derived. From decision-making and production management perspectives, the system reliability is a performance indicator for evaluating the capability of the manufacturing system. The manufacturing manager could determine whether the capacity of the manufacturing system can satisfy the customers' orders/demand or not. The remainder of this paper is organised as follows. The problem description, model construction for the same machine failure rate (Model I) and algorithm to generate the lower boundary vector for d are proposed in Section 2. The model in which a different machine with a different failure rate (Model II) is extended in Section 3. Numerical examples are demonstrated in Section 4 to illustrate two proposed algorithms and how the system reliability may be calculated. A discussion of the algorithm and conclusions are provided in Section 5.

2. Model I: all machines with the same failure rate

2.1 *Problem description and assumption*

We focus on a flow-shop manufacturing network in which products are make-to-stock. That is, this manufacturing network is a high-volume system with standardised machines and processes to produce identical or highly similar products. Products produced by this manufacturing network are then provided to customers from finished products stock (Stevenson 2007). This manufacturing network is requested to produce d units of products under the situation that WIP may be defective during processing. That is, there should be more input raw materials I than the output products O. To satisfy the demand, we first determine the amount of input raw materials and then evaluate the system reliability that satisfies the demand constraint. From a decision-making and production-management perspective, the system reliability can be a performance indicator for measuring the capability of the manufacturing network to determine whether it can fulfil customers' orders/demand or not. To evaluate the system reliability of the manufacturing network, this paper is studied based on the following assumptions:

(1) Each node (inspection station) is perfectly reliable. This assumption indicates that the inspection station would not damage any WIP/products, which implies that only the processing machines are considered.
(2) The capacity x_i of each arc a_i (machine) is a random variable according to a given probability distribution. In particular, no specified probability distribution is required for applying the proposed algorithm.
(3) The capacities of different arcs (machines) are statistically independent. In physical attribute each machine is an independent individual.
(4) No assembling action is taken for producing the product, and one material can produce one product. An assembling flow-shop system can be modelled by multiple lines based on the proposed models.
(5) Each defective WIP is reworked only once by the same machine. This implies that such a defective WIP is repaired until a usable state. If the defective WIP after reworking is still defective, it means that such a defective WIP is unrepeatable. Then, it is scrapped.

2.2 *Determination of input units*

In order to produce sufficient products for satisfying demand d, the input amount of raw materials should be pre-determined based on the failure rate of each machine. However, since we focus mainly on the output amount of this manufacturing network, the probability that a machine operates successfully is utilised for the following model construction. We name such a probability the success rate, where the success rate is (1–failure rate). Suppose that I units of raw materials are able to produce O units of products; we attempt to obtain the relationship between I and O fulfilling $O \geq d$. Take the simplest case with only one machine, for instance (Figure 1). I units of input raw

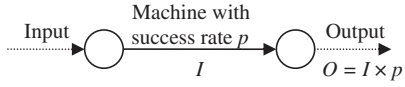


Figure 1. Input and output flow of a machine.

materials can produce $O = I \times p$ units of products, where p is the success rate, and $0 \leq p \leq 1$. For the single machine, the relationship between I and O can be derived in terms of the success rate p. That is, once the input I is determined, the output O can be easily obtained by multiplying by p.

We further extend the above concept to the plural machines case where each machine has the same success rate p. Let n denote the number of machines and a_i denote the ith machine. A manufacturing network with four ($n = 4$) machines (say a_1, a_2, a_3, and a_4) is an elementary model where each machine has the same success rate p. This elementary model includes a reworking action to re-input defective WIP to a previous machine (from a_3 back to a_2) considering a previous process (a_1) and a follow-up process (a_4). In this manufacturing system, I units of raw materials are processed through a_1, a_2, a_3, and a_4 sequentially (Figure 2). In Figure 2, the two black nodes represent the input inspection (check for raw materials) and output inspection (check for final products), respectively. White nodes between two machines denote the inspection for WIP to check if it can enter the next process (success) or should be scrapped (failure). The meshed node means that defective WIP (output from a_3) inspected by this inspection station can be reworked. That is, there would be I units of raw materials entering a_1 and Ip units of WIP outputting from a_1. Subsequently, Ip units of WIP enter a_2 and Ip^2 units of WIP output from a_2. Without reworking, we finally have $O = Ip^4$ units of product output from a_4 since the raw materials are processed through four machines with the same success rate p. To satisfy $O \geq d$ in this case, the input raw materials should be $I \geq d/p^4$ (recall that $O = Ip^4$). So far, the reworking action has not yet been considered.

Given that defective WIP output from rth machine can be reworked starting from previous k machine (i.e. starting from (r–k)th machine), for the special case that k is set to be zero this implies that the defective WIP is reworked at the same machine (rth machine). In this example, defective WIP output from machine a_3 ($r = 3$) is reworked starting from the previous one ($k = 1$) machine a_2 (also see Figure 2). This implies that Ip^2q units of defective WIP from a_3 would be re-inputted to a_2. However, the manufacturing process shown in Figure 2 is not distinguishable for the input flows of general manufacturing process and the reworking process. A revised topology of the manufacturing network as shown in Figure 3 would be able to describe the general manufacturing process and the reworking processes more clearly. The revised topology would be helpful for further analysis. Figure 3 also shows the input flow (under each arc) for each machine.

In terms of the concept of path, a dummy-machine, say a_2', is set to denote the machine a_2 doing the reworking action (Figure 3). Therefore, Ip^2q units of defective WIP from a_3 would be re-inputted to a_2', where q is the failure rate, and $q = 1 - p$. Through the reworking process, the amount of WIP output from a_2' is Ip^3q, and then it is re-inputted to a_3' (the dummy-machine for a_3). Thus, the manufacturing network finally has Ip^5q units of products output from a_4'. With the reworking process, the total output is $O = (Ip^4 + Ip^5q)$ from both a_4 and a_4', where the first term Ip^4 is produced by the general manufacturing process, while the second term Ip^5q is processed through the

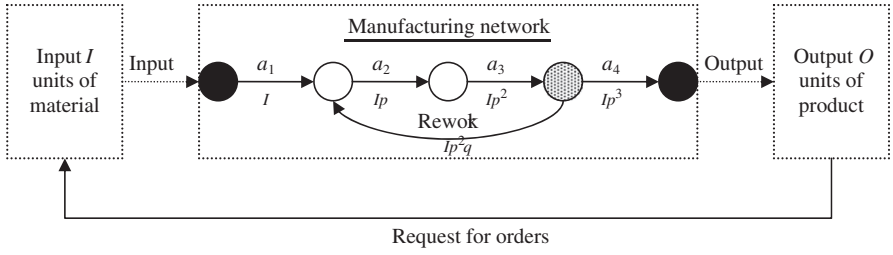


Figure 2. Manufacturing system with four machines.

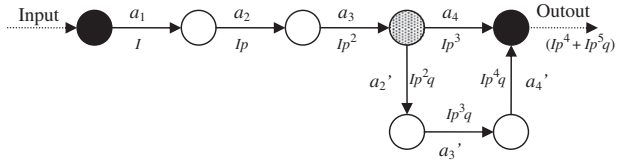


Figure 3. Revised topology for Figure 2.

reworking process. For satisfying $O \geq d$ in reworking case, the input amount should be set as

$$I \geq d/(p^4 + p^5 q). \tag{1}$$

Based on the above examples, we generalise the output amount from the manufacturing network. Given the demand d, number of machine n, and success rate p, the following lemma shows the relationship between input materials I and output products O in terms of d, n and p.

Lemma: *Assume that defective WIP output from the rth machine* $(1 < r \leq n)$ *is reworked starting from the previous k machine*(s) $(k \geq 0)$. *Then, I units of raw materials would produce* (i) Ip^n *units of product from general manufacturing process and* (ii) $Ip^{n+k}q$ *units of product from reworking process.*

The above lemma shows that the input determination is irrelevant with r. We thus have the output products $O = (Ip^n + Ip^{n+k}q)$, and it is necessary that $I \geq d/(p^n + p^{n+k}q)$ for obtaining sufficient output O that satisfies the demand d. The following equation guarantees the manufacturing network can produce sufficient output O that satisfies demand d.

$$I = d/(p^n + p^{n+k}q). \tag{2}$$

2.3 *Decomposition of the manufacturing network*

To analyse the manufacturing network in terms of paths, the manufacturing network in Figure 3 can be decomposed into two sets of paths, named the general processing path $P^{(G)}$ and the reworking path $P^{(R|r,r-k)} = P^{(R|3,3-1)}$, where $(R|3,3-1)$ denotes that defective WIP output from a_3 $(r = 3)$ is reworked starting from the previous one $(k = 1)$ machine (i.e. starting from a_2). Take the same example in Section 2.2, for instance; the set $\{a_1, a_2, a_3, a_4\}$ would be a general processing path $P^{(G)}$ (Figure 4). On the other side, a processing path with reworking action is $\{a_1, a_2, a_3, a_2', a_3', a_4'\}$. In fact, no defective WIP would be processed by a_1, a_2, and a_3, and the input flow would be zero for these arcs. Thus, a_1, a_2, and a_3 can be ignored, and merely the machines a_2', a_3', and a_4' doing the reworking action would be retained. Since the machine a_i and the dummy-machine a_i' are the same, the reworking path $P^{(R|3,2)}$ would be $\{a_2, a_3, a_4\}$ (Figure 5).

In order to extend the manufacturing network for n-machine cases, we have the following definition to decompose the manufacturing network into the general processing path $P^{(G)}$ and the reworking path $P^{(R|r,r-k)}$.

Definition: Assume that defective WIP output from the rth machine is reworked starting from the previous k machine(s). Given the general processing path $P^{(G)} = \{a_1, a_2, \ldots, a_n\}$, we can define that the reworking path $P^{(R|r,r-k)} = \{a_{r-k}, a_{r-k+1}, \ldots, a_n\}$.

To distinguish the sources of input flow, we have the input flow for each machine on the paths (either general processing path or reworking path) as follows:

$$f_i^{(G)} = Ip^{i-1} \text{ for } i \text{ such that } a_i \in P^{(G)} \tag{3}$$

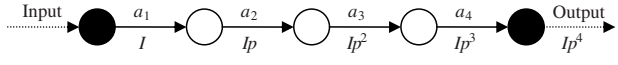


Figure 4. General processing path $P^{(G)}$ for Figure 2.

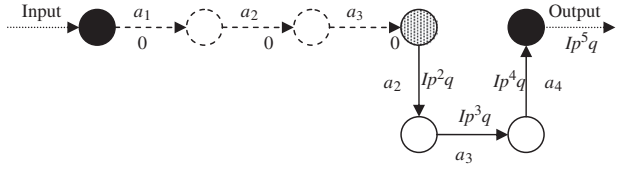


Figure 5. Reworking path $P^{(R|3,2)}$ for Figure 2.

and

$$f_i^{(\mathrm{R}|r,r-k)} = Ip^{i+k-1}q \text{ for } i \text{ such that } a_i \in P^{(\mathrm{R}|r,r-k)}, \tag{4}$$

where $f_i^{(\mathrm{G})}$ is the input flow for $a_i \in P^{(\mathrm{G})}$, and $f_i^{(\mathrm{R}|r,r-k)}$ is the input flow for $a_i \in P^{(\mathrm{R}|r,r-k)}$. Both Equations (3) and (4) represent the input flows that would be processed through each machine a_i, where Equation (3) is for the general processing path $P^{(\mathrm{G})}$, while Equation (4) is for the reworking path $P^{(\mathrm{R}|r,r-k)}$. Since the amount of input raw materials and the input flow for each machine are determined, it is possible to assess the capability if each machine can process the input WIP or not. Furthermore, we can derive the system reliability base on the decomposed paths and input flows.

2.4 *Determination of machine capacity and evaluation of system reliability*

The input raw materials/WIP processed by the ith machine a_i should satisfy the following constraints:

$$f_i^{(\mathrm{G})} + f_i^{(\mathrm{R}|r,r-k)} \le M_i. \tag{5}$$

Constraint (5) ensures that the general processing flow and the reworking flow do not exceed the maximal capacity M_i that each machine can provide. The term $f_i^{(\mathrm{G})} + f_i^{(\mathrm{R}|r,r-k)}$ is further defined as the loading of each machine, say l_i, and we have the following equation:

$$l_i = f_i^{(\mathrm{G})} + f_i^{(\mathrm{R}|r,r-k)}. \tag{6}$$

According to assumption 2, the capacity x_i of each machine a_i is a random variable and thus the manufacturing network is stochastic. Set c_i to be the number of possible capacities of a_i, and x_{ij} the jth possible capacity of a_i, where $j = 1, 2, \ldots, c_i$. Thus, x_i takes possible values $0 = x_{i1} < x_{i2} < \ldots < x_{ic_i} = M_i$. Under the state $X = (x_1, x_2, \ldots, x_n)$, constraint (7) is necessary to guarantee that a_i can process the input raw materials/WIP:

$$x_i \ge l_i = f_i^{(\mathrm{G})} + f_i^{(\mathrm{R}|r,r-k)} \text{ for } i = 1, 2, \ldots, n. \tag{7}$$

Given the demand d, the system reliability R_d is the probability that the output product from the manufacturing network is not less than d. Thus, the system reliability is $\Pr\{X|V(X) \ge d\}$, where $V(X)$ is defined as the maximum output under X. This implies that each machine should provide enough capacity to process the input raw materials/WIP and finally produce enough units of output products O. However, it is not wise to find all X such that $V(X) \ge d$ and then cumulate their probability to derived R_d. The minimal capacity vector Y in the set $\{X|V(X) \ge d\}$ is claimed to be the lower boundary vector for d. That is, Y is the lower boundary vector for d if and only if (i) $V(Y) \ge d$ and (ii) $V(Y') < d$ for any capacity vectors Y' such that $Y' < Y$. For the lower boundary vector $Y = (y_1, y_2, \ldots, y_n)$, we can calculate the system reliability as $R_d = \Pr\{X|X \ge Y\} = \Pr\{X|X \ge (y_1, y_2, \ldots, y_n)\} = \Pr\{x_1 \ge y_1\} \times \Pr\{x_2 \ge y_2\} \times \cdots \times \Pr\{x_n \ge y_n\}$.

2.5 *Algorithm I: for Model I to generate the lower boundary vector for d*

Algorithm I is utilised for Model I considering the same failure rate for all machines. Given that $P^{(\mathrm{G})} = \{a_1, a_2, \ldots, a_n\}$ and $P^{(\mathrm{R}|r,r-k)} = \{a_{r-k}, a_{r-k+1}, \ldots, a_n\}$, the lower boundary vector for d can be derived as the following steps.

Step 1: Determine the amount of input materials by Equation (8):

$$I = d/(p^n + p^{n+k}q). \tag{8}$$

Step 2: Determine the input flows for each machine by applying Equations (9)–(10):

$$f_i^{(\mathrm{G})} = Ip^{i-1} \text{ for } i \text{ such that } a_i \in P^{(\mathrm{G})}, \text{and} \tag{9}$$

$$f_i^{(\mathrm{R}|r,r-k)} = Ip^{i+k-1}q \text{ for } i \text{ such that } a_i \in P^{(\mathrm{R}|r,r-k)}. \tag{10}$$

Then, check the amount of input flows fulfils the following constraint:

$$f_i^{(\mathrm{G})} + f_i^{(\mathrm{R}|r,r-k)} \le M_i. \tag{11}$$

Step 3: Transform input flows from general processing path and reworking path into machines' loading vector $L=(l_1, l_2, \ldots, l_n)$ via

$$l_i = f_i^{(\mathrm{G})} + f_i^{(R|r,r-k)}. \tag{12}$$

Step 4: For each machine, find the smallest possible capacity such that $x_{ij} \ge l_i > x_{i(j-1)}$. Then, $Y=(y_1, y_2, \ldots, y_n)$ is the lower boundary vector for d where $y_i = x_{ij}$ for all i.

The exact amount of output products from this manufacturing network is $O=(Ip^n + Ip^{n+k}q)$, where I is determined by Step 1. The state Y determined from Step 4 is the lower boundary vector for d. The following theorem shows that the capacity Y generated from the algorithm is the lower boundary vector for d.

Theorem: *The capacity Y generated from the algorithm is the lower boundary vector for d.*

3. Model II: a different machine with a different failure rate

3.1 *Determination of input flow for each machine*

Based on Model I proposed in Section 2, we further consider the case of different machines with different failure rates. With I units of input raw materials, the output products produced by general processing path through machines $a_1, a_2, \ldots, a_n$ with success rates $p_1, p_2, \ldots, p_n$, respectively, would be $I\prod_{t=1}^{n} p_t$. Set the failure rate for machine a_i to be $q_i = 1 - p_i$; the output products from the reworking path is derived as $I\prod_{t=1}^{r-1} p_t \cdot q_r \prod_{t=r-k}^{n} p_t$, in which we assume that defective WIP output from the rth machine ($1 < r \le n$) are reworked starting from the previous k machine(s). Eventually, the total products output from both general processing path and reworking path would be $O = I\prod_{t=1}^{n} p_t + I\prod_{t=1}^{r-1} p_t \cdot q_r \prod_{t=r-k}^{n} p_t$.

To ensure that output products $O \ge d$, it is necessary that $I \ge d/(\prod_{t=1}^{n} p_t + \prod_{t=1}^{r-1} p_t \cdot q_r \prod_{t=r-k}^{n} p_t)$. Furthermore, we have the following equation to guarantee that a sufficient output O produced from I would satisfy demand d:

$$I = d \Bigg/ \left(\prod_{t=1}^{n} p_t + \prod_{t=1}^{r-1} p_t \cdot q_r \prod_{t=r-k}^{n} p_t \right). \tag{13}$$

Once I is determined, the extended model that different machine a_i with different success rate p_i would be conducted easily. To distinguish the sources of input flow, we have the input flow for each machine on the paths as follows:

$$f_i^{(\mathrm{G})} = \left(I\prod_{t=1}^{i} p_t \right) \Big/ p_i = I\prod_{t=1}^{i-1} p_t \text{ for } i \text{ such that } a_i \in P^{(\mathrm{G})}, \text{and} \tag{14}$$

$$f_i^{(\mathrm{R}|r,r-k)} = \left(I\prod_{t=1}^{r-1} p_t \cdot q_r \prod_{t=r-k}^{i} p_t \right) \Big/ p_i = I\prod_{t=1}^{r-1} p_t \cdot q_r \prod_{t=r-k}^{i-1} p_t \text{ for } i \text{ such that } a_i \in P^{(\mathrm{R}|r,r-k)} \tag{15}$$

The input flows processed by the ith machine a_i should satisfy the following constraints:

$$f_i^{(\mathrm{G})} + f_i^{(\mathrm{R}|r,r-k)} \le M_i, \tag{16}$$

and the loading of each machine is:

$$l_i = f_i^{(\mathrm{G})} + f_i^{(\mathrm{R}|r,r-k)}, \quad \text{for } i = 1, 2, \ldots, n. \tag{17}$$

Under the state $X=(x_1, x_2, \ldots, x_n)$, Equation (18) is necessary to guarantee that a_i can process the input raw materials/WIP:

$$x_i \ge l_i = f_i^{(\mathrm{G})} + f_i^{(\mathrm{R}|r,r-k)} \quad \text{for } i = 1, 2, \ldots, n. \tag{18}$$

3.2 *Algorithm II: for Model II to generate the lower boundary vector for* d

Algorithm II considers a different failure rate case for Model II. Given that $P^{(G)}=\{a_1, a_2, \ldots, a_n\}$ and $P^{(R|r,r-k)}=\{a_{r-k}, a_{r-k+1}, \ldots, a_n\}$, the lower boundary vector for d can be derived from the following steps.

Step 1: Determine the amount of input materials by Equation (19):

$$I = d \Big/ \left(\prod_{t=1}^{n} p_t + \prod_{t=1}^{r-1} p_t \cdot q_r \prod_{t=r-k}^{n} p_t \right). \tag{19}$$

Step 2: Determine the input flows for each machine by applying Equations (20)–(21):

$$f_i^{(G)} = \left(I \prod_{t=1}^{i-1} p_t \right) \text{ for } i \text{ such that } a_i \in P^{(G)}, \text{and} \tag{20}$$

$$f_i^{(R|r,r-k)} = \left(I \prod_{t=1}^{r-1} p_t \cdot q_r \prod_{t=r-k}^{i-1} p_t \right) \text{ for } i \text{ such that } a_i \in P^{(R|r,r-k)} \tag{21}$$

Then, check that the amount of input flows fulfils the following constraint,

$$f_i^{(G)} + f_i^{(R|r,r-k)} \le M_i. \tag{22}$$

Step 3: Transform input flows from general processing path and reworking path into the machines' loading vector $L=(l_1, l_2, \ldots, l_n)$ via

$$l_i = f_i^{(G)} + f_i^{(R|r,r-k)}. \tag{23}$$

Step 4: For each machine, find the smallest possible capacity such that $x_{ij} \ge l_i > x_{i(j-1)}$. Then, $Y=(y_1, y_2, \ldots, y_n)$ is the lower boundary vector for d where $y_i = x_{ij}$ for all i.

The exact amount of output products from this manufacturing network is $O = I\prod_{t=1}^{i} p_t + I\prod_{t=1}^{r-1} p_t \cdot q_r \prod_{t=r-k}^{i} p_t)$, where I is determined by Step 1. The state Y determined from Step 4 is the lower boundary vector for d.

4. Numerical examples

4.1 *Example 1: a tile-manufacturing network*

A tile-manufacturing network producing two product types is utilised to demonstrate Algorithms I and II. The tile manufacturing network is a high-volume production of only a certain type of product. The tiles (output products) are made starting from clay (input raw materials), and the tile-manufacturing processes can be characterised as three steps, say pressing (a_1), glazing, and burning (a_5). For the glazing step, different machines put the colours (by a_2), dust (by a_3), and smalts (by a_4) on the WIP (Andrés *et al.* 2005). In the tile-manufacturing network, defective WIP (smalts are put unevenly) output from a_4 ($r=4$) is reworked, starting from the previous one ($k=1$) machine (i.e. starting from a_3). Thus, the tile-manufacturing network can be described as a flow-shop manufacturing network with five machines ($n=5$) as shown in Figure 6. The tile-manufacturing network produces two types of product, say

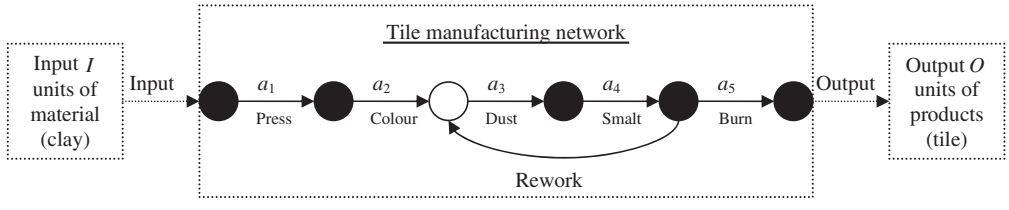


Figure 6. Tile manufacturing network with five machines.

products A and B, and the production routine is shown in Table 1. For both product types, this tile-manufacturing network has to satisfy the demand required as $d = 150$ per unit time.

4.1.1 *Case 1: machines with the same failure rate*

Case 1 is utilised to demonstrate Algorithm I proposed for the model in which all machines have the same failure rate (and the same success rate). Given the same success rate $p = 0.95$ for each machine and the capacity distribution of each machine in Table 2, we can evaluate the system reliability by Model I in the following procedure. First, we consider product A whose production processes go through all machines.

To analyse the manufacturing network in terms of paths, it is divided into the general processing path $P^{(G)} = \{a_1, a_2, a_3, a_4, a_5\}$ and the reworking path $P^{(R|r,r-k)} = P^{(R|4,4-1)} = \{a_3, a_4, a_5\}$. Then, we can generate the lower boundary vector for $d = 150$ as follows:

Step 1: Determine the amount of input materials: $I = d/(p^n + p^{n+k}q) = 150/(0.95^5 + 0.95^{5+1} \times 0.05) = 185.063$.

Step 2: Determine the input flows for each machine by applying Equations (9) and (10). The input flows of each machine are shown in rows 2–4 in Table 3. Since a_1 and a_2 are not on the reworking path, the input flow $f_1^{(R|4,3)} = f_2^{(R|4,3)} = 0$.

Table 1. Production routine of the tile manufacturing network in Example 1.

	Production routine				
Product type	a_1	a_2	a_3	a_4	a_5
A	1[a]	2	3	4	5
B	1	–	2	3	4

Note: [a]a_1 is the first processing machine to produce product A.

Table 2. Machine data for Example 1.

Machine (a_i)	Success rate[a] (p_i)	Capacity[b]	Probability	Machine (a_i)	Success rate[a] (p_i)	Capacity[b]	Probability
a_1	0.92	0	0.001	a_4	0.96	0	0.001
		50	0.002			40	0.002
		100	0.002			80	0.002
		150	0.005			120	0.005
		200	0.005			160	0.015
		250	0.985			200	0.975
a_2	0.93	0	0.001	a_5	0.90	0	0.005
		50	0.002			40	0.005
		100	0.002			80	0.010
		150	0.005			120	0.010
		200	0.010			160	0.010
		250	0.980			200	0.960
a_3	0.97	0	0.002				
		40	0.003				
		80	0.005				
		120	0.005				
		160	0.010				
		200	0.975				

Notes: [a]This column is only for Case 2.
[b]Units of raw materials/WIP that each machine can process per unit time.

Step 3: Transform input flows from general processing path and reworking path into machines' loading vector $L=(l_1, l_2, l_3, l_4, l_5)$ via

$$l_1 = f_1^{(G)} + f_1^{(R|4,3)} = 185.063 + 0 = 185.063;$$

$$l_2 = f_2^{(G)} + f_2^{(R|4,3)} = 175.810 + 0 = 175.810;$$

$$l_3 = f_3^{(G)} + f_3^{(R|4,3)} = 167.019 + 7.933 = 174.953;$$

$$l_4 = f_4^{(G)} + f_4^{(R|4,3)} = 158.668 + 7.537 = 166.205;$$

$$l_5 = f_5^{(G)} + f_5^{(R|4,3)} = 150.735 + 7.160 = 157.895.$$

Step 4: Find the minimal capacity vector $Y \geq L$ and thus $Y=(y_1, y_2, y_3, y_4, y_5)=(200, 200, 200, 200, 160)$.

The exact amount of output products from this manufacturing network is $O=(Ip^n + Ip^{n+k}q)=(19.525\times 0.95^5 + 19.525\times 0.95^{5+1}\times 0.05)=150$. The result after transformation is $Y=(200, 200, 200, 200, 160)$. After obtaining the lower boundary vector X, the system reliability $R_{150}^{(A)}$ for product A can be derived as follows:

$$\begin{aligned} R_{150}^{(A)} &= \Pr\{O \geq d\} = \Pr\{O \geq 150\} \\ &= \Pr\{X | X \geq (200, 200, 200, 200, 160)\} \\ &= \Pr\{x_1 \geq 200\} \times \Pr\{x_2 \geq 200\} \times \Pr\{x_3 \geq 200\} \times \Pr\{x_4 \geq 200\} \times \Pr\{x_5 \geq 160\} \\ &= 0.99 \times 0.99 \times 0.975 \times 0.975 \times 0.97 = 0.90376. \end{aligned}$$

However, colouring on a_2 is not necessary for product B, since the preference is for this product type to remain a natural colour. Hence, a_2 is removed from the manufacturing network for such product types. The manufacturing network is therefore revised as the general processing path $P^{(G)}=\{a_1, a_3, a_4, a_5\}$ and the reworking path $P^{(R|r,r-k)}=\{a_3, a_4, a_5\}$. We can easily solve this case by mapping to the four machines example shown in Figure 2. Without putting colours on tiles, the system reliabilities for product B is $R_{150}^{(B)}=0.91289$.

4.1.2 *Case 2: machines with different failure rates*

Case 2 is utilised to demonstrate Algorithm II proposed for the extended model where a different machine has a different failure rate. For both product types, the manufacturing network shown in Figure 6 still has to satisfy the demand $d=150$ per unit time, but now with a different success rate p_i for each machine in this case. Given the success rate p_i and capacity distribution of each machine in the same Table 2, we can evaluate the system reliability by Model II in the following procedure. For product A, the lower boundary vector for $d=150$ is generated as follows:

Step 1: Determine the amount of input materials: $I=d/(\prod_{t=1}^{n} p_t + \prod_{t=1}^{r-1} p_t \cdot q_r \prod_{t=r-k}^{n} p_t = 201.374$.

Step 2: Determine the input flows for each machine by applying equations (20) and (21). The input flows of each machine are shown in rows 2 to 4 in Table 4.

Table 3. Results of Step 2 for Case 1.

	i				
Input flow	1	2	3	4	5
$f_i^{(G)}=Ip^{i-1}$	185.063	175.810	167.019	158.668	150.735
$f_i^{(R\|4,2)}=Ip^{i+k-1}q$	0.000	0.000	7.933	7.537	7.160
$f_i^{(G)}+f_i^{(R\|4,2)}$	185.063	175.810	174.953	166.205	157.895

Step 3: Transform input flows from general processing path and reworking path into machines' loading vector $L=(l_1, l_2, l_3, l_4, l_5)=(201.374, 185.264, 178.981, 173.611, 166.667)$

Step 4: Find the minimal capacity vector $Y \geq L$ and thus $Y=(y_1, y_2, y, y_4, y_5)=(250, 200, 200, 200, 200)$.

The result after transformation is $Y=(250, 200, 200, 200, 200)$. After obtaining the lower boundary vector Y, the system reliability $R_{150}^{(A)}$ can be derived as follows:

$$\begin{aligned} R_{150}^{(A)} &= \Pr\{O \geq d\} = \Pr\{O \geq 150\} \\ &= \Pr\{X | X \geq (250, 200, 200, 200, 200)\} \\ &= \Pr\{x_1 \geq 250\} \times \Pr\{x_2 \geq 200\} \times \Pr\{x_3 \geq 200\} \times \Pr\{x_4 \geq 200\} \times \Pr\{x_5 \geq 200\} \\ &= 0.985 \times 0.99 \times 0.975 \times 0.975 \times 0.96 = 0.88992. \end{aligned}$$

Similarly, we have the system reliability $R_{150}^{(B)}=0.90347$ for product B in the different machine-failure-rates case.

4.1.3 *Case 3: parallel lines producing different product types*

In Case 3, we consider parallel production lines in the manufacturing network to show how the proposed model can be applied for a complicated system. In this case, two product types are processed on two parallel production lines, say P_1 and P_2, respectively. The manufacturing network has the general processing path $P_1^{(G)}=\{a_1, a_2, a_3, a_4, a_5\}$ and the reworking path $P_1^{(R|r, r-k)}=\{a_3, a_4, a_5\}$ for product A, while $P_2^{(G)}=\{a_6, a_7, a_8, a_9\}$ and the $P_2^{(R|r, r-k)}=\{a_7, a_8, a_9\}$ for product B. The manufacturing network with two parallel lines is shown in Figure 7. Suppose that a_6, a_7, a_8, and a_9 do the same operation as a_1, a_3, a_4, and a_5, respectively. For the same machine-failure-rates case, the capacity distribution of each machine is the same as that shown in Table 2.

In fact, we have solved this case in Case 1 since we derived the reliability for each production process. Thus, the system reliability can be calculated by $0.90376 \times 0.91289 = 0.82503$. This example shows that for a parallel-lines

Table 4. Results of Step 2 for Case 2.

Input flow	i				
	1	2	3	4	5
$f_i^{(G)}=I\prod_{t=1}^{i-1}p_t$	201.374	185.264	172.296	167.127	160.442
$f_i^{(R|4,2)}=I\prod_{t=1}^{r-1}p_t q_r \prod_{t=r-k}^{i-1}p_t$	0.000	0.000	6.685	6.485	6.225
$f_i^{(G)}+f_i^{(R|4,2)}$	201.374	185.264	178.981	173.611	166.667

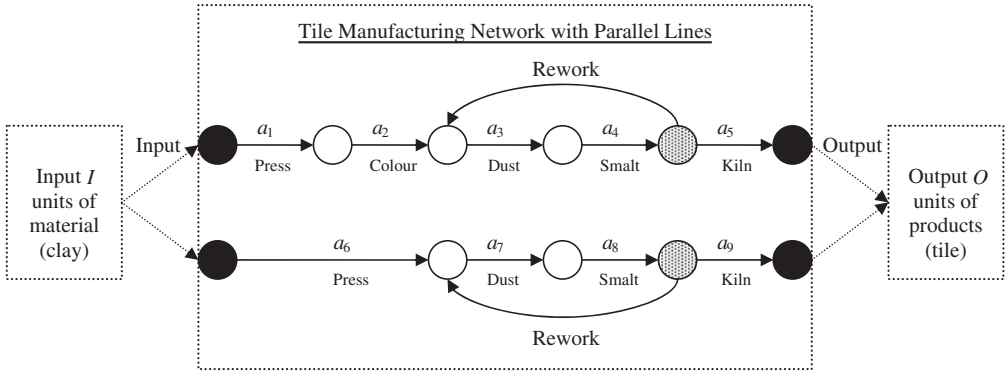


Figure 7. Tile manufacturing network with nine machines in two parallel lines.

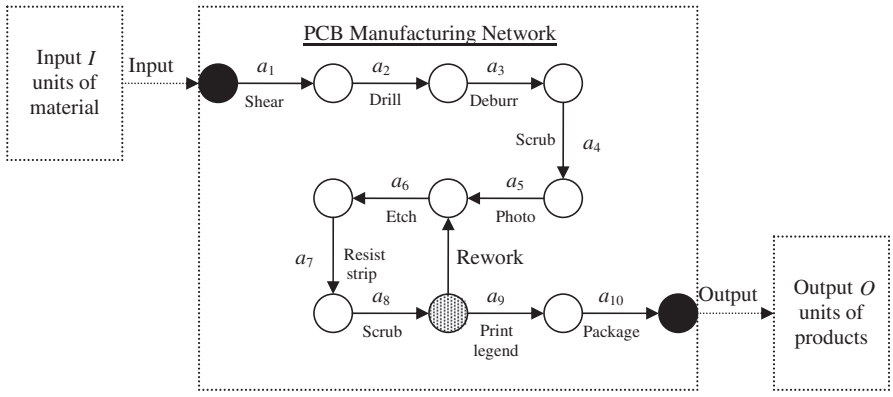


Figure 8. PCB manufacturing network with 10 machines.

manufacturing network, it can be decomposed into several production paths. According to the proposed model in this paper, we can easily deal with other complicated cases by transformation of the manufacturing network and decomposition method.

4.2 *Example 2: a PCB manufacturing network*

In Example 2, a typical printed circuit board (PCB) production process is utilised to demonstrate a manufacturing network with more processing machines. For the single-sided board production, the input raw material is a board with a thin layer of copper foil. For different product types, the production processes and sequences may be different. The general production process starts from shearing (a_1) in which the board is cut to a specific size. Subsequently, an automated drilling machine (a_2) drills holes through the board for mounting electronic components. After drilling, the deburring machine (a_3) removes copper particles from the board, and then the scrubbing machine (a_4) cleans the board. Following cleaning, the photo-imaging machine (a_5) creates the circuit pattern on the board. Using chemical etching (a_6) and resist stripping (a_7), the copper not part of the circuit pattern is removed from the board. Once again, the scrubbing machine (a_8) is used to clean the chemicals and resist on the board. After scrubbing, the legend printing process (a_9) prints the required logos or letters. Finally, the production process is completed with the packaging stage (a_{10}). Thus, the PCB manufacturing network is described as a flow-shop manufacturing network with 10 machines ($n=10$), as shown in Figure 8.

In the PCB manufacturing network, some defective WIP output from a_8 ($r=8$) is reworked starting from the previous two ($k=2$) machines (i.e. starting from a_6). To analyse the PCB manufacturing network in terms of paths, it is divided into the general processing path $P^{(G)}=\{a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9, a_{10}\}$ and the reworking path $P^{(R|r,r-k)}=P^{(R|8,8-2)}=\{a_6, a_7, a_8, a_9, a_{10}\}$. Given the success rate p_i and capacity distribution of each machine in Table 5, the system reliability for $d=200$ is $R_{200}=0.92350$.

5. Discussion and conclusion

5.1 *Discussion*

An integer issue for the algorithms is first discussed in this section. The algorithms proposed in this paper are generalised in terms of real numbers, since we measure the input flow that each machine processes per unit time. That is, from the flow perspective, the amounts of raw materials/WIP/products are averaged by time, and thus the value is a real number. Besides, the machine failure rate is always a real number, and thus the output flow (WIP/products) processed by each machine is indeed a real number. However, eventually the exact amount of output products would be an integer that can be obtained by (flow $\times$ time period) after a long-term time period. If the equations and constraints are modified by utilising the $\lfloor x \rfloor$ or $\lceil x \rceil$ operations to solve integer cases (where $\lfloor x \rfloor$ is the largest integer such that $\lfloor x \rfloor \leq x$ and $\lceil x \rceil$ is the smallest integer such that $\lceil x \rceil \geq x$), the procedure in the algorithm

Table 5. Machine data for Example 2.

Machine (a_i)	Success rate (p_i)	Capacity	Probability	Machine (a_i)	Success rate (p_i)	Capacity	Probability
a_1	0.98			a_6	0.95	0	0.002
		0	0.005			120	0.003
		300	0.005			240	0.005
		600	0.990			360	0.015
						480	0.975
a_2	0.96			a_7	0.96	0	0.002
		0	0.001			60	0.002
		300	0.001			120	0.005
		600	0.998			180	0.005
						240	0.010
						300	0.976
a_3	0.99			a_8	0.99	0	0.001
		0	0.002			60	0.001
		250	0.003			120	0.001
		500	0.995			180	0.012
						240	0.015
						300	0.970
a_4	0.99	0	0.005	a_9	0.98	0	0.002
		250	0.010			120	0.003
		500	0.985			240	0.995
a_5	0.96	0	0.001	a_{10}	0.99		
		120	0.001			0	0.010
		240	0.002			120	0.015
		360	0.003			240	0.975
		480	0.993				

would be very complicated to calculate. For example, given the input amount I and machine success rate p, the output from the first machine would be $\lfloor Ip \rfloor$, the output from the second machine would be $\lfloor \lfloor Ip \rfloor \times p \rfloor$, and so on. Taking a numerical case for explanation, let input $I = 100$ and success rate $p = 0.95$ for each machine. Thus, the output of this manufacturing process is $O = \lfloor \lfloor \lfloor \lfloor 100 \times 0.95 \rfloor \times 0.95 \rfloor \times 0.95 \rfloor \times 0.95 \rfloor = 80$. However, if we take the real number to calculate the output, the output should be $100 \times 0.95^4 = 81.45$. However, after a 100-unit time period, the output of integer case is underestimated to be 8000, while the real output should be 8145. Moreover, this paper intends to determine the amount of input raw materials in terms of success rate p before evaluating the system reliability. That is, if there are four machines in the manufacturing network, the output products would be Ip^4, and this implies that the overall success rate is p^4. Given the demand d, we can easily obtain $I \geq d/p^4$ for satisfying the demand in terms of the overall success rate. If we consider the integer case, the output would be $\lfloor \lfloor \lfloor \lfloor Ip \rfloor \times p \rfloor \times p \rfloor \times p \rfloor$, and it is not intuitive to determine the amount of raw materials input. Once the reworking process is considered, the algorithm would be increasingly complicated.

The other issue to be addressed is that the defective WIP is reworked by specific reworking machines which are not allocated in the general processing path. That is, this manufacturing network has additional machines for reworking rather than the original ones. Again, take the tile-manufacturing network (Example 1) for instance; suppose that the defective WIP output from a_4 can be reworked starting from a reworking machine that does the same operation as a_3, and the follow-up machines are still a_4 and a_5. To model such a case, let $\hat{a}_3$ be the reworking machine that implements the same activity as a_3; we thus have a general processing path $P^{(G)} = \{a_1, a_2, a_3, a_4, a_5\}$ and a reworking path $P^{(R|r,r-k)} = \{\hat{a}_3, a_4, a_5\}$. Notice that, there is no more defective WIP being re-inputted to machine a_3, and the input flow of a_3 is only $f_3^{(G)}$ (recall that the original amount is $f_3^{(G)} + f_3^{(R|4,3)}$). The defective WIP amount, $f_3^{(R|4,3)}$, is now going to the reworking machine $\hat{a}_3$. The other input flows are not changed, and thus the loading of other machines is the same. By assuming that $\hat{a}_3$ has the same capacity probability distribution as a_3, the system reliabilities for $d = 150$ are 0.90195 for the same machine failure rate case and 0.88814 for the different machine failure rates case.

5.2 *Conclusion*

To assess the robustness of the manufacturing system, this paper first constructed a stochastic-flow network model considering reworking action with the same failure rate for all machines (Model I). Then, we extended this to a different failure rate case (Model II). In addition, we attempted to evaluate the capability whether a manufacturing system can satisfy the demand d or not. Thus, the probability that a manufacturing network could produce d units of demand with reworking action, named system reliability, is evaluated as a performance indicator to assess its capability. However, in a manufacturing network, the input raw materials/WIP would not equal the amount of output, since some would be reworked or be scrapped when being processed by each machine. Thus, in contrast to the situation in previous studies (Xue 1985, Lin *et al*. 1995, Lin 1998, 2003, 2004, 2007a, b, 2009, 2010, Zuo *et al*. 2007, Yeh 2008), the network would break the flow conservation. To analyse the input flows, a manufacturing network is decomposed into a general processing path (without reworking) and several reworking paths. Based on these paths, we determine the capacity that each machine should provide for eventually satisfying the amount of output products d. An algorithm is further utilised to generate the lower boundary vector such that each machine has sufficient capacity to produce enough products satisfying demand d. The proposed model and algorithm can be extended to more than one reworking action case intuitively. Based on the system reliability, the manufacturing manager could conduct the sensitivity analysis to investigate the most important machine in a manufacturing network to make the system more reliable.

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